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Original Article

The profiles of miRNAs and lncRNAs in peripheral blood neutrophils exosomes of diffuse cutaneous systemic sclerosis

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ABSTRACT

Background: Diffuse cutaneous systemic sclerosis (dSSc) is a systemic autoimmune disease with skin fibrosis. Neutrophils display important roles in autoimmunity, inflammation, vasculopathy and fibrosis. Exosomes (EXOs) are cell-derived vesicles contained various noncoding RNAs, mRNA and proteins with biological roles.
Objective: To investigate the roles of miRNAs and lncRNAs from dSSc neutrophils EXOs.
Methods: EXOs were isolated from cultured neutrophils supernatants and identified by transmission electron microscopy. Global expression of miRNAs and lncRNAs in neutrophils EXOs were sequenced by Illumina HiSeq 3000 and bioinformatic analyses were performed by R/Bioconductor. Genes were validated by real-time quantitative PCR.
Results: In profiles of neutrophils EXOs, we identified 22 dysregulated miRNAs and 281 dysregulated lncRNAs. Predicted target genes of them were enriched in GO, KEGG and Reactome pathways, Wnt, AMPK, IL-23 and NOTCH signaling pathways were selected for further analysis. Widely interactions among them were also found. Human dermal microvascular endothelial cells and human primary skin fibroblasts were stimulated with dSSc neutrophils EXOs, these fibrosis related genes were detected and some changes were found, such as ENST00000533886.1-hsa-miR-1268a-CAMK2G in Wnt and IL-23 signaling pathways, ENST00000610091.1-hsa-miR-299-3p, 512-3p-CPT1A in IL-23 and AMPK signaling pathways, NR_001564.2, ENST00000520562.1, ENST00000596567.1-hsa-miR-299-3p, 512-3p -TFDP2 in IL-23, AMPK and NOTCH signaling pathways.
Conclusions: The profiles of miRNAs and lncRNAs of neutrophils EXOs provided novel clues for dSSc pathogenesis. We identified several gene pairs in the Wnt, AMPK, IL-23 and NOTCH signaling pathways, which could be potential biomarkers and therapeutic targets in dSSc.
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1. Introduction

[1] Systemic sclerosis (SSc) is a multisystem connective tissue disease with an unknown aetiology that is characterized by autoimmune disorders, inflammation, vasculopathy and extensive fibrosis in the skin and other organs [1]. Based on the extension of skin fibrosis, SSc can be classified into two clinical subsets: limited cutaneous SSc (lSSc, distal fibrosis to the elbow and the knee joint, including the face) and diffuse cutaneous SSc (dSSc, proximal fibrosis). Fibrosis is the hallmark of SSc and contributes to its mortality, which is the highest among connective tissue diseases [2]. The molecular mechanisms of fibrosis in SSc are complicated, and many pathways have important roles, including transforming growth factor-β (TGF-β) signaling, canonical Wnt signaling, Toll-like receptor 4 signaling, reactive oxygen species (ROS) generation, adenosine monophosphate-activated protein kinase (AMPK) signaling, NOTCH signaling, interleukin (IL)-6 and IL-23, and other factors [1,3].

Neutrophils, which are specialist inflammatory and secretory cells of the innate immune system, play a critical role during the pathology of SSc [4]. During the innate immune response in SSc, transmigrating neutrophils could both damage and activate endothelial cells directly and then lead to vascular remodelling

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[1] List of abbreviations: SSc: systemic sclerosis; lSSc: limited cutaneous SSc; dSSc: diffuse cutaneous SSc; TGF-β: transforming growth factor-β; ROS: reactive oxygen species; AMPK: adenosine monophosphate-activated protein kinase; IL: interleukin; EXOs: exosomes; miRNAs: microRNAs; lncRNAs: long noncoding RNAs; NCs: normal controls; neutrophils EXOs: EXOs from neutrophils cultured supernatants; RT-qPCR: real-time quantitative PCR; DLS: dynamic light scattering; TEM: transmission electron microscopy; GO: Gene Ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes; HDMECs: human dermal microvascular endothelial cells; DMEM: Dulbecco's modified Eagle's medium; RQ: relative quantity; relative quantity; ECM: extracellular matrix.

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and inflammation by enforcing a self-sustaining positive feedback loop, in addition to the overall outcome of the inflammatory response [5]. Meanwhile, neutrophils can be triggered to release exosomes (EXOs), proteases, cytokines and ROS extracellularly, leading to dysregulation of oxidative homeostasis, and modification of proteins, lipids, DNAs and RNAs [4]. According to previous studies, neutrophils in the blood of SSc patients play activating roles in autophagy and vasculopathy, most likely due to platelet-derived microparticle-associated HMGB1 or ROS-promoted extracellular HMGB1 [6,7].

EXOs are cell-derived extracellular vesicles, defined as cup-shaped spherical microvesicles 30–150 nm in size, with specific surface markers such as tumour susceptibility gene 101, integrins and tetraspanins, including CD63, CD9, and CD81/82. EXOs can be secreted by almost all types of cells and are present in all biological fluids. EXOs contain high levels of various RNAs, DNAs and proteins [8], and EXOs derived from different cells or under different biological conditions may have different compositions. EXOs widely participate in intercellular signaling, such as information transfer, diffuse inflammation, antigen presentation, tumour growth, metastasis, and angiogenesis [9]. EXOs have been widely studied in a variety of diseases. For example, EXO-derived microRNAs (miRNAs) act as predictive biomarkers for diffuse large B-cell lymphoma chemotherapy resistance [10], and EXO-miRNA-423-3p derived from cardiac fibroblasts mediates the cardioprotective effects of ischaemic post-conditioning [11]. Meanwhile, neutrophils EXOs can enable the innate immune response in an autocrine and paracrine fashion [12].

Recently, many studies have shown that epigenetic alterations may play a more important role than hereditary susceptibility in a wide range of developmental and physiological cellular processes in SSc. In the current studies [13,14], miRNAs (<50 nucleotides) and long noncoding RNAs (lncRNAs, >200 nucleotides) are most widely studied in SSc pathogenesis.

Emerging information has revealed that neutrophils EXOs contain many characteristic miRNAs and lncRNAs involved in the pathogenesis of SSc [15,16]. In this study, we characterized the miRNA and lncRNA profiles in neutrophils EXOs of SSc patients and healthy individuals using RNA sequencing analysis and investigated the pathogenesis, therapeutic targets and biomarkers of SSc patients.

2. Materials and methods

2.1. Patients and controls

Whole blood samples were obtained from 23 dSSc patients and 18 healthy individuals from Xiangya Hospital of Central South University in this study. All dSSc patients and normal controls (NCs) were Han Chinese. Patients fulfilled the ACR/EULAR 2013 classification criteria for diffuse cutaneous SSc [17]. Among them, 5 dSSc patients and 5 age- and gender-matched NCs were selected for the neutrophils EXO miRNA and lncRNA gene sequencing analysis, all 23 dSSc patients and 18 NCs were included for validation by real-time quantitative PCR (RT-qPCR) in different cell models. In the 5 dSSc patients, 2 patients were anti-scl70 (+) and 3 patients were anti-scl70 (-), all the 5 SSc patients were anti-RNP (-). The clinical characteristics of dSSc patients and NCs are shown in the supplementary table S1.

2.2. Neutrophils purification and culture

For neutrophils purification, 10 ml peripheral whole blood samples were collected in preservative-free EDTA anticoagulant tubes from dSSc patients and NCs and processed within 2 h. We extracted neutrophils with Histopaque-1077 and Histopaque-1119 (Sigma-Aldrich, USA) according to the manufacturer's instructions. The cell pellets were used for culture or stored at −80 °C until analysis.

For neutrophils culture, pellets were resuspended in the appropriate volume (2–5 × 10^6 cell/ml) of 1640 media (Gibco, USA) supplemented with 10% EXO-free foetal bovine serum (Gibco, USA, prepared by obtaining the supernatant after centrifugation at 150,000xg for 3 h at 4 °C). Neutrophils were cultured for 2 h at 37 °C in a 5% CO_2 humidified incubator. The cell pellets and supernatants were collected separately for further analysis.

2.3. EXO isolation and identification

We isolated EXOs from all neutrophils cultured supernatants (20−30 ml/sample) (neutrophils EXOs) using an exoEasy Maxi Kit (Qiagen, Frederick, USA) according to the manufacturer's instructions. The EXOs were finally eluted using 600 μl Buffer XE and stored at −80 °C until analysis. EXOs were identified through size distribution by using dynamic light scattering (DLS) on a Zetasizer Nano ZS (Malvern Instrument, UK) and through morphology by using transmission electron microscopy (TEM) (FEI Tecnai G2 Spirit Twin, Czech Republic) via negative staining. We also performed bead-based flow cytometry with EXOome-Human CD63 Isolation/Detection Dynabeads (Invitrogen, Thermo Fisher Scientific, Lithuania) and anti-human CD63-PE (12-0639, eBioscience, USA) according to the manufacturer's protocol. We used a FACSCalibur flow cytometer (BD Biosciences, USA) and FlowJo software (TreeStar) for analysis.

2.4. Library construction and RNA sequencing analysis

Total RNA was extracted from neutrophils EXOs using TRIzol (Invitrogen Life Technologies, CA, USA) according to the manufacturer's protocol. For preparation of RNA libraries, the NEBNext® Multiplex Small RNA Library Prep Set for miRNAs and NEBNext® Ultra™ RNA Library Prep Kit for lncRNAs from Illumina (NEB, USA) were used according to the manufacturer's instructions. The quantity and integrity of total RNA and purified library products were both assessed by separately using the Agilent 2200 TapeStation (Agilent Technologies, USA) and Qubit®2.0 (Life Technologies, USA). The paired-end sequenced (PE150) with an Illumina HiSeq 3000 platform and the sequencing data analysis were performed at Guangzhou RiboBio Co., Ltd. (Guangzhou, China). The bioconductor package edgeR and DESeq2 were used for differential expression analyses of the RNASeq read counts. EdgeR was used for calculating differentially expressed miRNAs according to the criteria of fold change>1.2 and adjusted p-value <0.05. DESeq2 was used for calculating differentially expressed lncRNAs according to the criteria of fold change>1.5 and adjusted p-value <0.05.

2.5. Bioinformatic analysis

TargetScan, miRDB, miRTarBase and miRWalk databases were used to predict target genes of the selected miRNAs, and only target genes which were predicted by at least 3 databases were chosen. As lncRNAs regulate target genes in two major categories, in cis and in trans, the potential target genes regulated in cis of lncRNAs were obtained by integrating the differentially expressed lncRNA and its adjacent (10 kb) differentially expressed mRNA data. For in trans regulation, sequences of differentially expressed lncRNAs and mRNAs were extracted, BLAST software was used for initial screening (e<1E-5), and RNAplex software was used for rescreening to identify potential target genes of lncRNAs. The target genes of differentially expressed miRNAs and lncRNAs were selected for Gene Ontology (GO), Kyoto Encyclopedia of Genes

and Genomes (KEGG) and Reactome pathway analyses by RclusterProfiler [18].

For the miRNA-lncRNA interaction prediction, we mainly used miRanda, PITA and RNAhybrid softwares to predict the recognition elements between miRNAs and lncRNAs, and we finally obtained predicted miRNAs according to the lncRNAs of interest. To show the interactions of differentially expressed miRNAs, lncRNAs and target mRNAs, we constructed coexpression networks of lncRNA-miRNA pairs and lncRNA-miRNA-mRNA pairs by Cytoscape software.

2.6. Cell culture and treatment

Human dermal microvascular endothelial cells (HDMECs) were purchased from CellBio (catalogue # CBR130858) [19]. Primary skin fibroblasts were obtained from NC biopsy samples from our previous studies [20]. HDMECs and fibroblasts were cultured in Dulbecco's modified Eagle's medium (DMEM, Gibco, USA) supplemented with 10% EXO-free foetal bovine serum (described above) at 37 °C in a 5% CO_2 humidified incubator. Both HDMECs and fibroblasts were used when they reached 80–90% confluence and were then stimulated with 20% neutrophils EXOs (volume concentration, 160ul neutrophil EXOs buffer and 640ul medium) of dSSc patients and NCs for 48 h in 24-well plates. Each independent experiment was performed three times with more than six samples in each group in duplicate.

2.7. Validation of gene expression by RT-qPCR

Total RNA was isolated from neutrophils, HDMECs and fibroblasts using TRIzol according to the manufacturer's instructions.

For lncRNAs and mRNAs, cDNAs were obtained from a PrimeScript™ RT reagent Kit (Perfect Real Time, TaKaRa, Japan). The relative expression was measured by gene-specific primers [shown in Supplementary table S2 and S3] with TB Green (TB Premix Ex Taq RT-PCR kit, TaKaRa, Japan) and the 7500 Fast Real-Time PCR system (Applied Biosystems, CA, USA). The housekeeping gene GAPDH was selected for normalizing and calculating the relative quantity (RQ) value of gene expression in each sample.

For miRNAs, a miDETECT A TrackTM miRNA qRT-PCR Starter Kit (RiboBio, China) was used to obtain cDNAs and measure the relative expression of miRNAs in the 7500 Fast Real-Time PCR system. The miRNA-specific primers and a uni-reversal primer were purchased from RiboBio. U6 was used as the endogenous control to normalize the sample data as previously described [19,20]. Each sample was tested in duplicate.

2.8. Statistical analysis

GraphPad Prism 5 was used for statistical analysis of the data except the bioinformatic data. An unpaired *t*-test was used for numerical variables with a normal distribution. The Wilcoxon rank sum test was used for data with a non-normal distribution. The data are shown as the mean ± SEM. $p < 0.05$ was considered significant difference.

3. Results

3.1. Identification of neutrophils EXOs

To verify successful isolation, we characterized neutrophils EXOs by TEM, bead-based flow cytometry and DLS. These neutrophils EXOs showed a typical cup-shaped morphology of approximately 30−150 nm in size and positive expression of the EXO-specific marker CD63 (Fig. 1A-C).

3.2. Identification of differentially expressed miRNAs in dSSc neutrophils EXOs

In the profile of neutrophils EXO miRNAs, we identified a total of 22 differentially expressed miRNAs between dSSc patients and NCs. Bioinformatic analysis indicated that the 12 upregulated miRNAs have 2708 target genes, and the 10 downregulated miRNAs have 2090 target genes (Fig. 2A-B) [shown in Supplementary file 1.1 and 1.2]. Among the 22 dysregulated miRNAs, we found that the expression of hsa-miR-218-5p and hsa-miR-144-3p were significantly different between anti-scl70 (-) and anti-scl70 (+) groups (Supplementary Fig. 1A).

For upregulated target genes, the GO annotation of gene biological process revealed that Wnt signaling pathway, TGF-β signaling pathway, actin filament organization and extracellular matrix (ECM) organization were enriched. KEGG pathway analysis suggested that the PI3K-Akt, MAPK, FoxO and mTOR signaling pathways were enriched. For downregulated target genes, GO analysis showed response to TGF-β signaling pathway, PDGF signaling pathway and regulation of ECM organization were significantly enriched. The AMPK, MAPK, Chemokine and FoxO signaling pathways were significantly enriched in KEGG analysis (Fig. 2C).

Based on the predicted target genes, 8 upregulated miRNAs and 51 miRNA-mRNA pairs participated in the Wnt signaling pathway (Fig. 2D). Additionally, 5 downregulated miRNAs and 33 miRNA-mRNA pairs participated in the AMPK signaling pathway (Fig. 2E) [shown in Supplementary file 1.3].

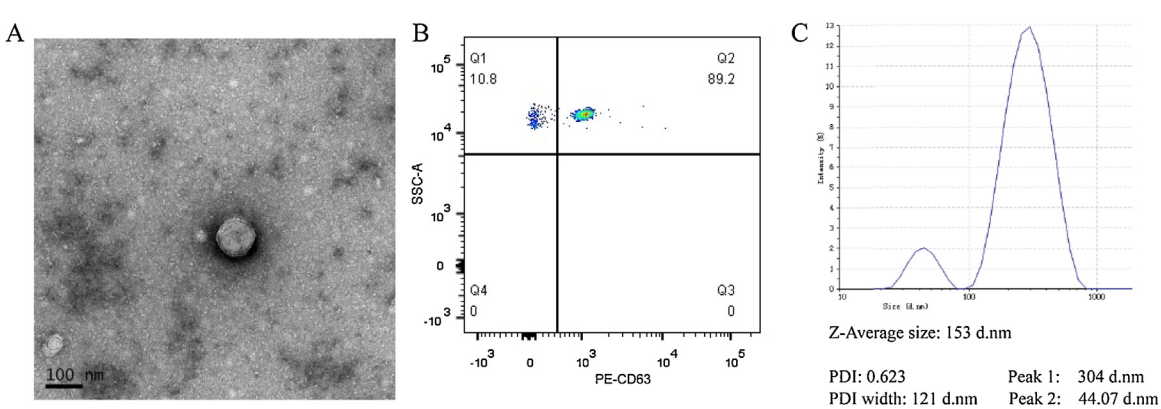

Fig. 1. Identification of neutrophils EXOs. (A) Neutrophils EXOs showed the typical cup shape with TEM (scale bars, 100 nm), (B) positive expression for CD63 in bead-based flow cytometry and (C) an average size of approximately 30-150 nm by DLS.

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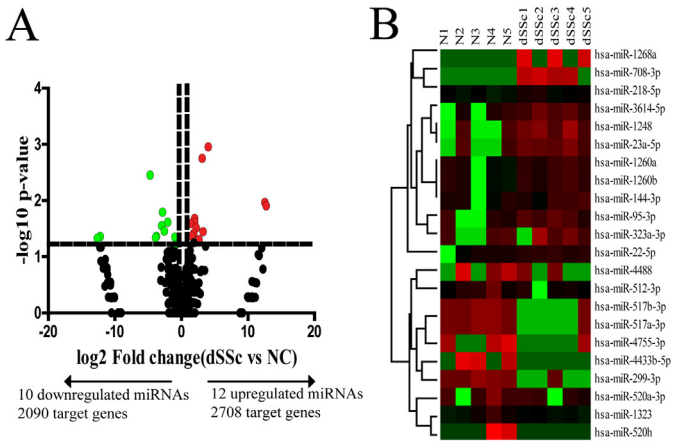

C

Target genes of Upregulated miRNAs

Wnt signaling pathway (GO)	3.91E-09
TGF-beta signaling pathway (GO)	2.87E-03
Actin filament organization (GO)	3.02E-04
Extracellular matrix organization (GO)	1.38E-03
Response to fibroblast growth factor (GO)	5.41E-04
Response to epidermal growth factor (GO)	7.56E-04
PI3K-Akt signaling pathway (KEGG)	1.14E-04
MAPK signaling pathway (KEGG)	9.41E-04
FoxO signaling pathway (KEGG)	2.03E-04
mTOR signaling pathway (KEGG)	1.21E-02

Target genes of Downregulated miRNAs

AMPK signaling pathway (KEGG)	8.20E-05
TGF-beta signaling pathway (GO)	2.28E-02
PDGF signaling pathway (GO)	8.00E-03
Regulation of extracellular matrix organization (GO)	3.54E-02
MAPK signaling pathway (KEGG)	1.08E-03
Chemokine signaling pathway (KEGG)	3.49E-02
FoxO signaling pathway (KEGG)	1.62E-02

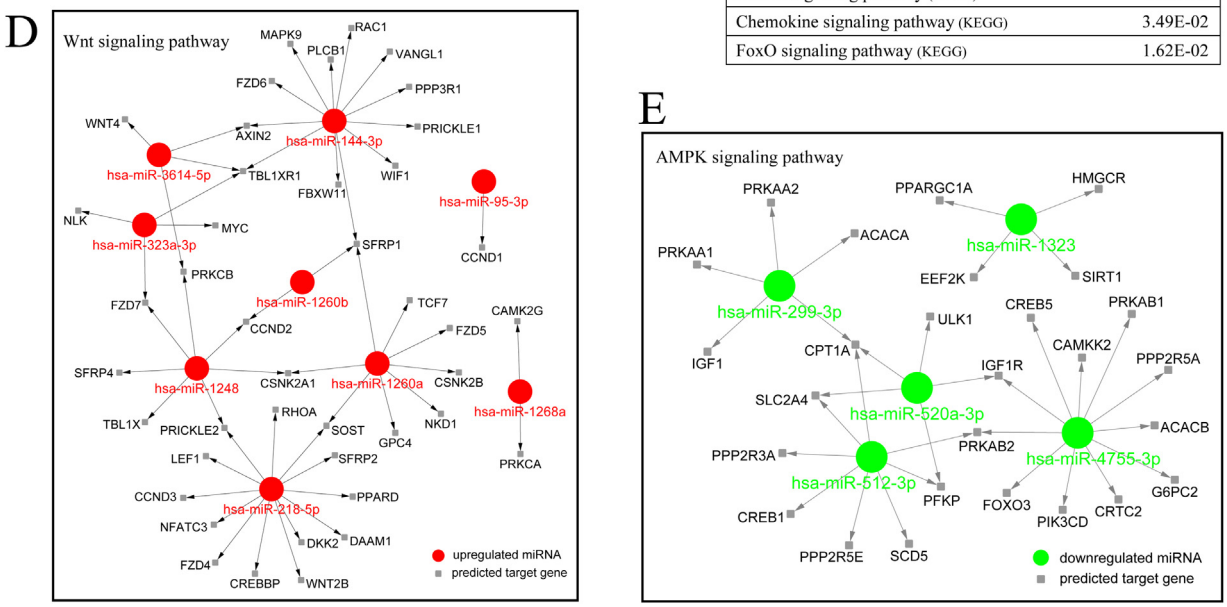

Fig. 2. Identification of differentially expressed miRNAs in dSSc neutrophils EXOs. The profile analysis showed 22 differentially expressed miRNAs between dSSc patients and NCs in (A) a volcano plot and (B) heatmap, (C) GO and KEGG analysis were performed for their target genes and showed many related biological process and signaling pathways. Among them, (D) 8 upregulated miRNAs and 51 miRNA-mRNA pairs participated in the Wnt signaling pathway, (E) 5 downregulated miRNAs and 33 miRNA-mRNA pairs participated in the AMPK signaling pathway.

3.3. Identification of differentially expressed lncRNAs in dSSc neutrophils EXOs

To obtain comprehensive gene information, we also investigated lncRNA profiling of neutrophils EXOs between dSSc patients and NCs. We identified 281 differentially expressed lncRNAs. The prediction analysis indicated that the 119 upregulated lncRNAs have 998 target genes, and the 162 downregulated lncRNAs have 1954 target genes (Fig. 3A-B) [shown in Supplementary file 2.1 and 2.2]. Among the 281 dysregulated lncRNAs, 23 lncRNAs were significantly different between anti-scl70 (-) and anti-scl70 (+) groups (Supplementary Fig. 1B).

GO analysis of upregulated target genes suggested that regulation of IL-6, chemokine and interferon-beta/alpha production were enriched. Reactome pathway analysis indicated that the IL-23 and IL-20 family signaling were significantly changed. Similarly for downregulated target genes, type I interferon process and cytokine production involved in immune response were enriched in biological processes of GO analysis. Meanwhile, NOTCH signaling pathway and ECM organization were enriched in Reactome pathway analysis (Fig. 3C).

In addition, 10 upregulated lncRNAs and 15 lncRNA-mRNA pairs were included in the IL-23 signaling pathway (Fig. 3D), 22 downregulated lncRNAs and 24 lncRNA-mRNA pairs were included in the NOTCH signaling pathway (Fig. 3E) [shown in Supplementary file 2.3].

3.4. Interaction analysis of differentially expressed lncRNAs-miRNAs in dSSc neutrophils EXOs

miRNAs and lncRNAs both play crucial roles in regulating cellular processes, and lncRNAs can regulate the expression of miRNAs. Here, we further analysed the lncRNA-miRNA interactions in dSSc neutrophils EXOs. Based on the differentially expressed lncRNAs and miRNAs, we matched these lncRNA-miRNA pairs with interaction relationships. The results suggested that a total of 16 miRNAs (7 downregulated and 9 upregulated miRNAs) and 71 lncRNAs (42 downregulated and 29 upregulated lncRNAs) exhibited an interaction relationship and generated 112 lncRNA-miRNA pairs (Fig. 4A) [shown in Supplementary file 3.1].

Furthermore, from the lncRNA-miRNA network, we selected specific lncRNAs and miRNAs, which had the same predicted target

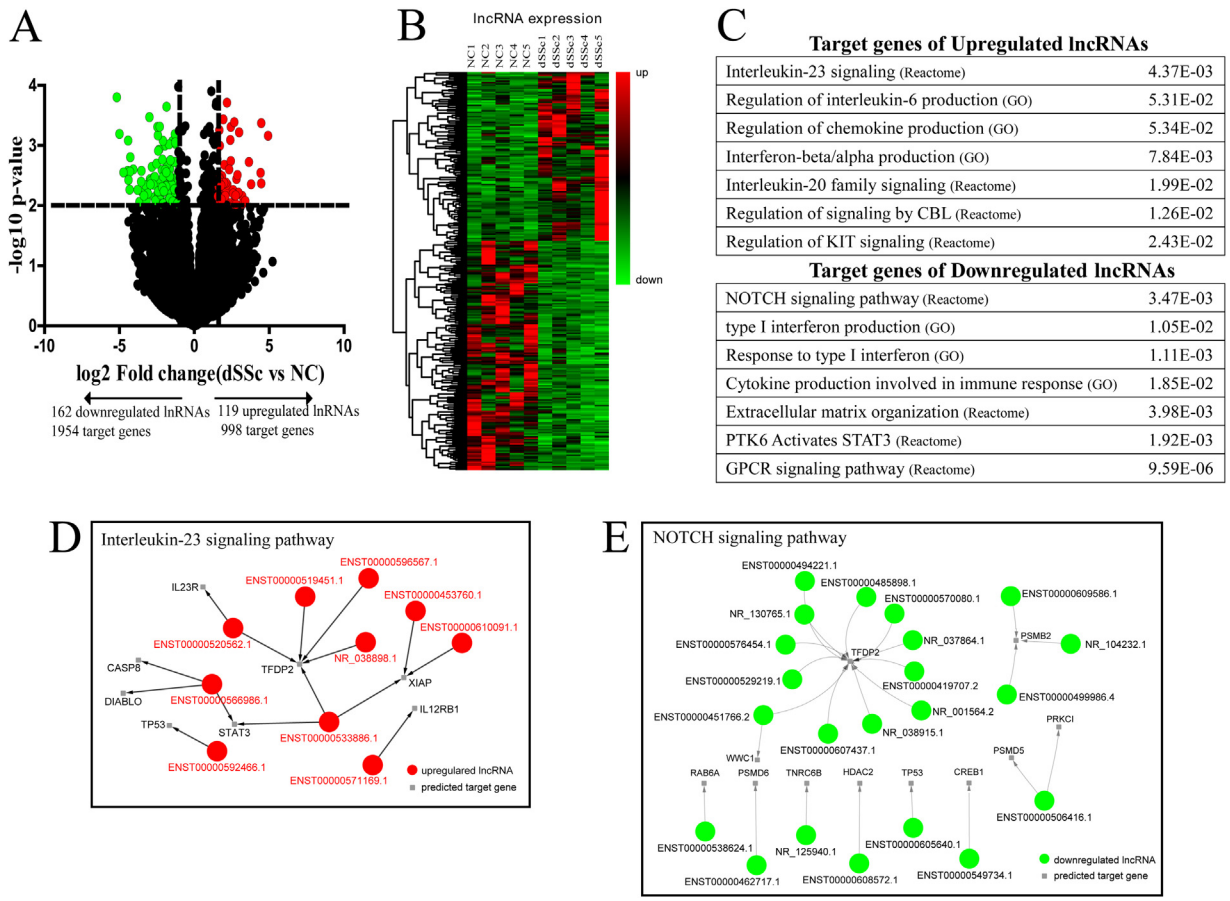

Fig. 3. Identification of differentially expressed lncRNAs in dSSc neutrophils EXOs. The profile analysis showed 281 differentially expressed lncRNAs between dSSc patients and NCs in (A) a volcano plot and (B) heatmap, (C) GO and Reactome analysis were performed for their target genes and showed many related biological process and signaling pathways. Among them, (D) 10 upregulated lncRNAs and 15 lncRNA-mRNA pairs were included in the IL-23 signaling pathway, (E) 22 downregulated lncRNAs and 24 lncRNA-mRNA pairs were included in the NOTCH signaling pathway.

mRNAs, to form a subnetwork comprising 9 miRNAs (3 downregulated and 6 upregulated miRNAs), 19 lncRNAs (13 downregulated and 6 upregulated lncRNAs), 30 miRNA-lncRNA pairs, 135 mRNAs and 187 lncRNA-mRNA or miRNA-mRNA pairs (Fig. 4B) [shown in Supplementary file 3.2].

3.5. Validation of differentially expressed miRNAs and target mRNAs in neutrophils of dSSc patients, HDMECs and fibroblasts after stimulation with dSSc neutrophils EXOs

To further confirm the pathways associated with differentially expressed miRNAs, lncRNAs and mRNAs of neutrophils EXOs, we validated the expression of these genes in neutrophils of dSSc patients and NCs, as well as in HDMECs and fibroblasts stimulated with 20 % neutrophils EXOs for 48 h, and detected the effect of expression of the predicted target genes.

For the upregulated differentially expressed miRNAs in the Wnt signaling pathway, all 8 miRNAs and the CAMK2G and PRKCA genes (target genes of has-miR-1268a) were validated by RT-qPCR. Most miRNAs and mRNAs were significantly differentially expressed in the three cell types, and many genes were significantly consistent with miRNA profile analysis (Fig. 5A, C, E). In particular, the expression of has-miR-1268a was elevated both in neutrophils and HDMECs (Fig. 5A, C). Conversely, the PRKCA gene was downregulated in neutrophils (Fig. 5B), identifying a miRNA-mRNA pair that negatively regulates each other in EXO-derived cells. Interestingly, we found that the expression levels of CAMK2G and PRKCA were increased in HDMECs and fibroblasts (Fig. 5D, F), which indicated that the Wnt signaling pathway in downstream target cells was activated by neutrophils EXOs.

For the downregulated miRNAs in the AMPK signaling pathway, the expression of all 5 miRNAs consistently declined in neutrophils EXO-derived cells (Fig. 5A). In the HDMECs and fibroblasts, the expression of these miRNAs was mostly elevated (Fig. 5C, E), similar to the upregulated expression of the IGF1R, PRKAB2, ULK1 and CPT1A genes (target genes of has-miR-299-3p, 520a-3p and 4755-3p) in the three cell types (Fig. 5B, D, F). These findings also indicated that the AMPK signaling pathway in these cells was widely activated by neutrophils EXOs.

3.6. Validation of differentially expressed lncRNAs and target mRNAs in neutrophils of dSSc patients, HDMECs and fibroblasts after stimulation with dSSc neutrophils EXOs

Meanwhile, the results showed that almost all of the selected IL-23 signaling pathways related lncRNAs were significantly upregulated in neutrophils and fibroblasts, and most lncRNAs were elevated in HDMECs (Fig. 6A, C, E). Similar to the target genes, IL23R (target gene of ENST00000520562.1) and TFDP2 (target gene of ENST00000596567.1) were both elevated in HDMECs and fibroblasts (Fig. 6D, F), and we obtained two confirmed lncRNA-mRNA pairs. The expression of the validated genes was nearly entirely matched with that of lncRNA sequencing analysis.

For the selected lncRNAs and mRNAs in the NOTCH signaling pathway, expression of the majority of genes was significantly upregulated in the three cell types (Fig. 6A-F). Therefore, the

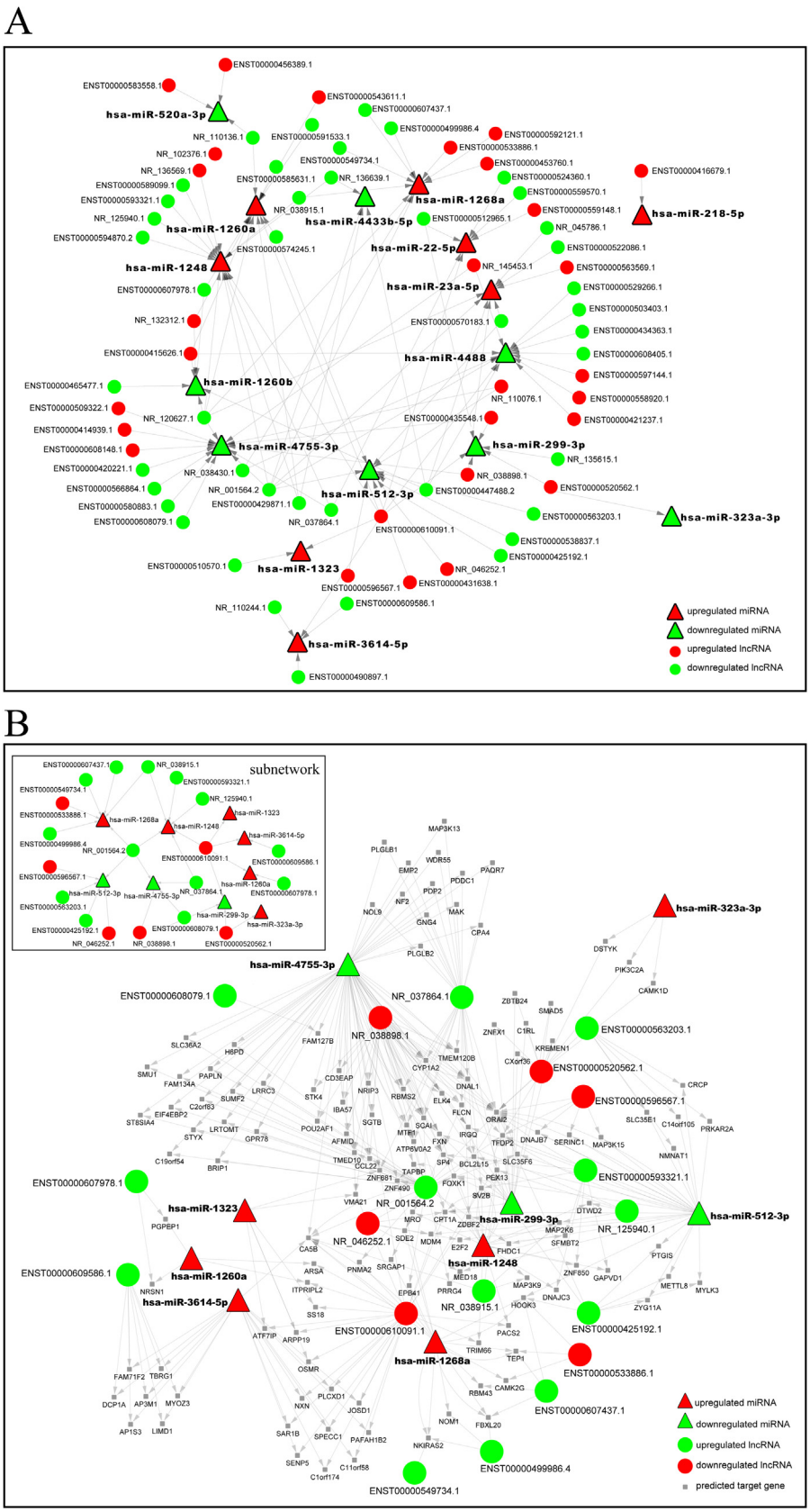

Fig. 4. Interaction analysis of differentially expressed lncRNAs-miRNAs in dSSc neutrophils EXOs. (A) A lncRNA-miRNA network with interaction relationship and differentiated expression. (B) A lncRNA-miRNA-mRNA network with interaction relationship, differentiated expression and co-prediction.

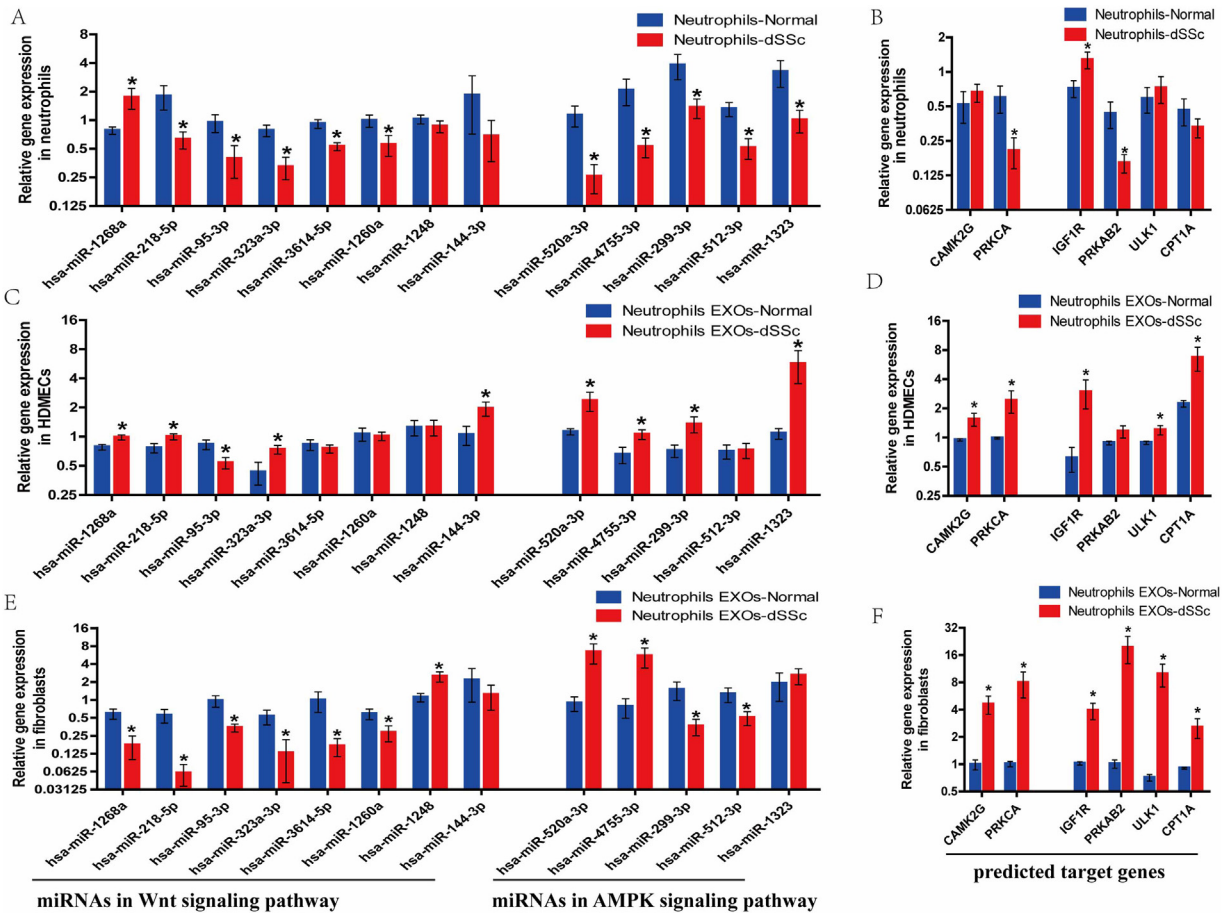

Fig. 5. Validation of differentially expressed miRNAs and target mRNAs in three cell models. The miRNAs in Wnt signaling pathway and AMPK signaling pathway, and selected target mRNAs were validated by RT-qPCR (A and B) in neutrophils of dSSc patients, (C and D) in HDMECs and (E and F) in fibroblasts after stimulation with dSSc neutrophils EXOs. (*dSSc vs NC).

ENST00000608572.1-HDAC2 pair might play an important role in the activation of the NOTCH signaling pathway in HDMECs and fibroblasts (Fig. 6C-F).

By matching the validated genes with the cotarget mRNA in the interaction network, we finally identified three mRNAs and three lncRNA-miRNA-mRNA pairs that were coactivated in two or three pathways, including the ENST00000533886.1-hsa-miR-1268a-CAMK2G pair in the Wnt signaling and IL-23 signaling pathways, the ENST00000610091.1-hsa-miR-299-3p, 512-3p-CPT1A pair in the IL-23 signaling and AMPK signaling pathways, the NR_001564.2, ENST00000520562.1, ENST00000596567.1 - hsa-miR-299-3p, 512-3p-TFDP2 pair in the IL-23 signaling, AMPK signaling and NOTCH signaling pathways.

4. Discussion

This is the first study to perform systematic analyses of neutrophils EXO miRNA and lncRNA expression profiles in dSSc. We acquired many signaling pathways related miRNAs, lncRNAs and mRNAs of dSSc neutrophils EXOs, including the Wnt, AMPK, IL-23 and NOTCH signaling pathways. Notably, according to the interaction analysis for the lncRNA-miRNA-mRNA subnetwork, we found three gene pairs that had been validated in vitro, revealing original and valuable information for SSc pathogenesis and treatment.

The first interacting and cotargeted lncRNA-miRNA-mRNA pair was the ENST00000533886.1-hsa-miR-1268a-CAMK2G, which was activated in both the Wnt signaling and IL-23 signaling pathways. In recent years, the canonical Wnt pathway has been characterized as an important mediator of sustained fibroblast activation in fibrotic diseases, eliciting fibrotic responses through β-catenin and TGF-β [21,22]. According to clinical trials targeting the Wnt/β-catenin pathway, inhibitors such as C-82, tankyrase and porcupine inhibitors are effective, well-tolerated and safe for long-term applications [23,24]. The canonical Wnt pathway also has multiple transcriptional target genes involved in fibrosis and in tissue repair. For example, Wnt3a might participate in the modulation of the endothelial to mesenchymal transition in HDMECs via the regulation of cadherin and vimentin mRNA expression [25]. In addition, in early research, the circulating level of IL-23 was decreased in SSc patients [26,27], and the IL-23R gene did not confer a risk for SSc [28,29]. However, in recent years, IL-23 had increasing evidence in certain fibrotic sites of SSc patients [30,31]. According to previous reports, IL23R is one of various SSc-associated genes that can enhance the proliferation of Th17 cells by regulating IL-23 activity [32], and the major alleles of the rs11209026 SNP and rs11465804 SNP of *IL23R* were found to be associated with susceptibility to anti-topoisomerase I antibody-positive and protective for pulmonary arterial hypertension in SSc patients [33]. A case study reported that ustekinumab, an IL-12/IL-23 inhibitor, successfully improved the skin tightening of SSc patients [34]. Our study supported above views and provided some novel miRNAs, lncRNAs and target genes of Wnt signaling and IL-23 signaling pathways for further study, which may become a useful tool to intervene in the progression of SSc.

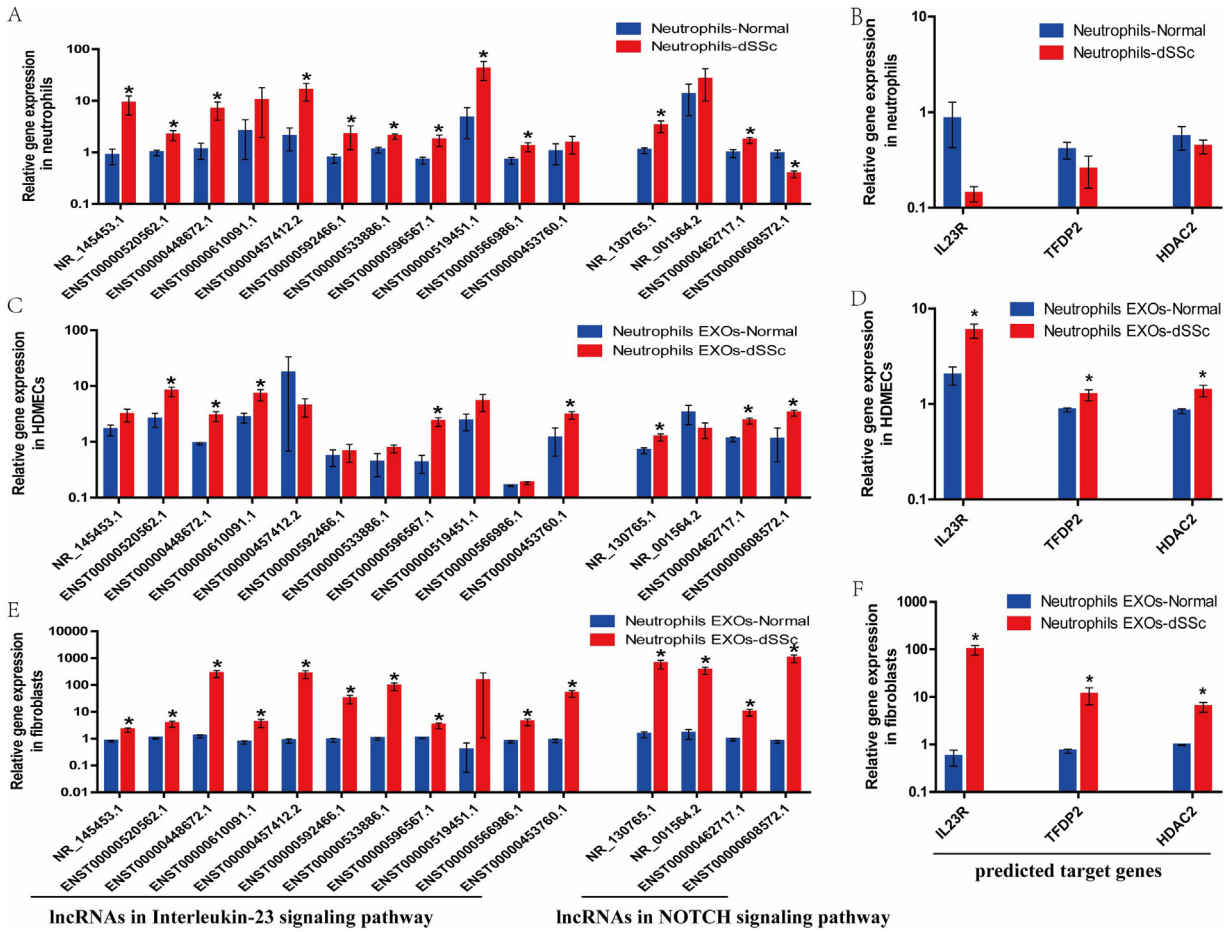

Fig. 6. Validation of differentially expressed lncRNAs and target mRNAs in three cell models. The lncRNAs in IL-23 signaling pathway and NOTCH signaling pathway, and selected target mRNAs were validated by RT-qPCR (A and B) in neutrophils of dSSc patients, (C and D) in HDMECs and (E and F) in fibroblasts after stimulation with dSSc neutrophils EXOs. (*dSSc vs NC).

Second, the ENST00000610091.1-hsa-miR-299-3p, 512-3p-CPT1A pair was involved in both the IL-23 signaling and AMPK signaling pathways. AMPK is mainly known as a conserved and ubiquitously expressed energy sensor (by sensing increases in the AMP:ATP and ADP:ATP ratios) in eukaryotic cells, and AMPK activity can control diverse aspects of cellular metabolism [35]. Recent studies have shown that AMPK-mediated signaling is mechanistically involved in autophagy regulation, through specific phosphorylation of autophagy-relevant proteins or by indirectly affecting the activity of additional autophagy regulators [36]. Furthermore, some studies have reported the negative effect of AMPK on TGF-β signaling and their opposing effects on several disease conditions, such as cancer, fibrosis, and angiogenesis [37]. However, no relevant reports have examined the correlation between AMPK signaling and SSc patients now [35].

In addition, the NR_001564.2, ENST00000520562.1, ENST 00000596567.1-hsa-miR-299-3p, 512-3p-TFDP2 pair was involved in the IL-23 signaling, AMPK signaling and NOTCH signaling pathways. NOTCH, Wnt and Hedgehog-signaling cascades are morphogen pathways that play crucial roles in development and tissue homeostasis [38]. Many studies have noted that NOTCH signaling is activated in SSc patients and plays an important role in fibrosis, and the inhibition of this pathway may represent a new treatment for fibrosis [39–41]. Expression of the target gene HDAC2 was significantly downregulated in SSc B cells and was negatively correlated with global histone H4 acetylation and skin thickness [42]. In our study, NOTCH signaling was activated in dSSc neutrophils EXOs and target cells, as well as the other three pathways mentioned.

5. Conclusion

EXOs are effective forms of transportation and may be the main source of epigenetic molecular markers or drug targets in the future [43]. In this study, the gene expression profiles and interactions of miRNAs and lncRNAs of peripheral blood neutrophils EXOs provided novel findings and clues about SSc pathogenesis, and the biological functions and four related signaling pathways require further in-depth studies. As a hypothesis generating study, we are looking for that, neutrophils EXO molecules may be potential disease biomarkers and therapeutic targets of angiogenesis and fibrosis in SSc.

Ethical approval & consent to participate

All blood samples were obtained with the written informed consent of patients. The study was approved by the Ethical Committee of Xiangya Hospital of Central South University No. 201303293, 201404360, 201212074.

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Declaration of Competing Interest

The authors have no conflict of interest to declare.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jdermsci.2020.02.009.

References

[1] X. Zuo, L. Zhang, H. Luo, Y. Li, H. Zhu, Systematic approach to understanding the pathogenesis of systemic sclerosis, Clin. Genet. 92 (2017) 365–371.
[2] L. Li, X. Zuo, H. Luo, Y. Li, Y. You, L. Duan, W. Zhang, H. Zhao, T. Li, W. Ning, Y. Xie, S. Liu, X. Xie, Y. Jiang, S. Wu, H. Zhu, Y. Zhou, Mortality trend of inpatients with connective tissue diseases: 2005-2014, Zhong Nan Da Xue Xue Bao Yi Xue Ban 42 (2017) 927–933.
[3] Y. Li, J. Huang, M. Guo, X. Zuo, MicroRNAs regulating signaling pathways: potential biomarkers in systemic sclerosis, Genomics Proteomics Bioinfo. 13 (2015) 234–241.
[4] L. Glennon-Alty, A.P. Hackett, E.A. Chapman, H.L. Wright, Neutrophils and redox stress in the pathogenesis of autoimmune disease, Free Radic. Biol. Med. 125 (2018) 25–35.
[5] N. Maugeri, P. Rovere-Querini, A.A. Manfredi, Disruption of a regulatory network consisting of neutrophils and platelets fosters persisting inflammation in rheumatic diseases, Front. Immunol. 7 (2016) 182.
[6] N. Maugeri, A. Capobianco, P. Rovere-Querini, G.A. Ramirez, E. Tombetti, P.D. Valle, A. Monno, V. D'Alberti, A.M. Gasparri, S. Franchini, A. D'Angelo, M.E. Bianchi, A.A. Manfredi, Platelet microparticles sustain autophagy-associated activation of neutrophils in systemic sclerosis, Sci. Transl. Med. 10 (2018).
[7] N. Maugeri, P. Rovere-Querini, M. Baldini, E. Baldissera, M.G. Sabbadini, M.E. Bianchi, A.A. Manfredi, Oxidative stress elicits platelet/leukocyte inflammatory interactions via HMGB1: a candidate for microvessel injury in sytemic sclerosis, Antioxid. Redox Signal. 20 (2014) 1060–1074.
[8] E. van der Pol, A.N. Boing, P. Harrison, A. Sturk, R. Nieuwland, Classification, functions, and clinical relevance of extracellular vesicles, Pharmacol. Rev. 64 (2012) 676–705.
[9] M. Yanez-Mo, P.R. Siljander, Z. Andreu, A.B. Zavec, F.E. Borras, E.I. Buzas, K. Buzas, E. Casal, F. Cappello, J. Carvalho, E. Colas, A. Cordeiro-da Silva, S. Fais, J.M. Falcon-Perez, I.M. Ghobrial, B. Giebel, M. Gimona, M. Graner, I. Gursel, M. Gursel, N.H. Heegaard, A. Hendrix, P. Kierulf, K. Kokubun, M. Kosanovic, V. Kralj-Iglic, E.M. Kramer-Albers, S. Laitinen, C. Lasser, T. Lener, E. Ligeti, A. Line, G. Lipps, A. Llorente, J. Lotvall, M. Mancek-Keber, A. Marcilla, M. Mittelbrunn, I. Nazarenko, E.N. Nolte-'t Hoen, T.A. Nyman, L. O'Driscoll, M. Olivan, C. Oliveira, E. Pallinger, H.A. Del Portillo, J. Reventos, M. Rigau, E. Rohde, M. Sammar, F. Sanchez-Madrid, N. Santarem, K. Schallmoser, M.S. Ostenfeld, W. Stoorvogel, R. Stukelj, S.G. Van der Grein, M.H. Vasconcelos, M.H. Wauben, O. De Wever, Biological properties of extracellular vesicles and their physiological functions, J. Extracell. Vesicles 4 (2015) 27066.
[10] Y. Feng, M. Zhong, S. Zeng, L. Wang, P. Liu, X. Xiao, Y. Liu, Exosome-derived miRNAs as predictive biomarkers for diffuse large B-cell lymphoma chemotherapy resistance, Epigenomics 11 (2019) 35–51.
[11] H. Luo, X. Li, T. Li, L. Zhao, J. He, L. Zha, Q. Qi, Z. Yu, microRNA-423-3p exosomes derived from cardiac fibroblasts mediates the cardioprotective effects of ischaemic post-conditioning, Cardiovasc. Res. 115 (2019) 1189–1204.
[12] J. Rossaint, K. Kuhne, J. Skupski, H. Van Aken, M.R. Looney, A. Hidalgo, A. Zarbock, Directed transport of neutrophil-derived extracellular vesicles enables platelet-mediated innate immune response, Nat. Commun. 7 (2016) 13464.
[13] H. Zhu, H. Luo, Y. Li, Y. Zhou, Y. Jiang, J. Chai, X. Xiao, Y. You, X. Zuo, MicroRNA-21 in scleroderma fibrosis and its function in TGF-beta-regulated fibrosis-related genes expression, J. Clin. Immunol. 33 (2013) 1100–1109.
[14] B. Zhou, H. Zhu, H. Luo, S. Gao, X. Dai, Y. Li, X. Zuo, MicroRNA-202-3p regulates scleroderma fibrosis by targeting matrix metalloproteinase 1, Biomed. Pharmacother. 87 (2017) 412–418.
[15] S. Juzenas, G. Venkatesh, M. Hubenthal, M.P. Hoeppner, Z.G. Du, M. Paulsen, P. Rosenstiel, P. Senger, M. Hofmann-Apitius, A. Keller, L. Kupcinskas, A. Franke, G. Hemmrich-Stanisak, A comprehensive, cell specific microRNA catalogue of human peripheral blood, Nucleic Acids Res. 45 (2017) 9290–9301.
[16] P.J. Wermuth, S. Piera-Velazquez, S.A. Jimenez, Exosomes isolated from serum of systemic sclerosis patients display alterations in their content of profibrotic and antifibrotic microRNA and induce a profibrotic phenotype in cultured normal dermal fibroblasts, Clin. Exp. Rheumatol. 35 (Suppl 106) (2017) 21–30.
[17] F. van den Hoogen, D. Khanna, J. Fransen, S.R. Johnson, M. Baron, A. Tyndall, M. Matucci-Cerinic, R.P. Naden, T.A. Medsger Jr., P.E. Carreira, G. Riemekasten, P.J. Clements, C.P. Denton, O. Distler, Y. Allanore, D.E. Furst, A. Gabrielli, M.D. Mayes, J.M. van Laar, J.R. Seibold, L. Czirjak, V.D. Steen, M. Inanc, O. Kowal-Bielecka, U. Muller-Ladner, G. Valentini, D.J. Veale, M.C. Vonk, U.A. Walker, L. Chung, D.H. Collier, M.E. Csuka, B.J. Fessler, S. Guiducci, A. Herrick, V.M. Hsu, S. Jimenez, B. Kahaleh, P.A. Merkel, S. Sierakowski, R.M. Silver, R.W. Simms, J. Varga, J.E. Pope, Classification criteria for systemic sclerosis: an American College of Rheumatology/European League against Rheumatism collaborative initiative, Arthritis Rheum. 65 (2013) (2013) 2737–2747.
[18] G. Yu, L.G. Wang, Y. Han, Q.Y. He, clusterProfiler: an R package for comparing biological themes among gene clusters, OMICS 16 (2012) 284–287.
[19] S. Gao, X. Zuo, D. Liu, Y. Xiao, H. Zhu, H. Zhang, H. Luo, The roles of neutrophil serine proteinases in idiopathic inflammatory myopathies, Arthritis Res. Ther. 20 (2018) 134.
[20] B. Zhou, X.X. Zuo, Y.S. Li, S.M. Gao, X.D. Dai, H.L. Zhu, H. Luo, Integration of microRNA and mRNA expression profiles in the skin of systemic sclerosis patients, Sci. Rep. 7 (2017) 42899.
[21] C. Bergmann, J.H. Distler, Canonical Wnt signaling in systemic sclerosis, Lab. Invest. 96 (2016) 151–155.
[22] B. Korman, Evolving insights into the cellular and molecular pathogenesis of fibrosis in systemic sclerosis, Transl. Res. 209 (2019) 77–89.
[23] R. Lafyatis, J.C. Mantero, J. Gordon, N. Kishore, M. Carns, H. Dittrich, R. Spiera, R. W. Simms, J. Varga, Inhibition of beta-catenin signaling in the skin rescues cutaneous adipogenesis in systemic sclerosis: a randomized, double-blind, placebo-controlled trial of C-82, J. Invest. Dermatol. 137 (2017) 2473–2483.
[24] C. Beyer, H. Reichert, H. Akan, T. Mallano, A. Schramm, C. Dees, K. Palumbo-Zerr, N.Y. Lin, A. Distler, K. Gelse, J. Varga, O. Distler, G. Schett, J.H. Distler, Blockade of canonical Wnt signaling ameliorates experimental dermal fibrosis, Ann. Rheum. Dis. 72 (2013) 1255–1258.
[25] W.J. Lee, J.H. Park, J.U. Shin, H. Noh, D.H. Lew, W.I. Yang, C.O. Yun, K.H. Lee, J.H. Lee, Endothelial-to-mesenchymal transition induced by Wnt 3a in keloid pathogenesis, Wound Repair Regen. 23 (2015) 435–442.
[26] A. Olewicz-Gawlik, A. Danczak-Pazdrowska, B. Kuznar-Kaminska, J. Gornowicz-Porowska, K. Katulska, D. Trzybulska, H. Batura-Gabryel, W. Silny, D. Poplawski, P. Hrycaj, Interleukin-17 and interleukin-23: importance in the pathogenesis of lung impairment in patients with systemic sclerosis, Int. J. Rheum. Dis. 17 (2014) 664–670.
[27] P. Gourh, F.C. Arnett, S. Assassi, F.K. Tan, M. Huang, L. Diekman, M.D. Mayes, J.D. Reveille, S.K. Agarwal, Plasma cytokine profiles in systemic sclerosis: associations with autoantibody subsets and clinical manifestations, Arthritis Res. Ther. 11 (2009) R147.
[28] B. Rueda, J. Broen, O. Torres, C. Simeon, N. Ortego-Centeno, M.M. Schrijvenaars, M.C. Vonk, V. Fonollosa, F.H. van den Hoogen, M.J. Coenen, J. Sanchez-Roman, M.A. Aguirre-Zamorano, R. Garcia-Portales, A. Pros, M.T. Camps, M.A. Gonzalez-Gay, J. Martin, T.R. Radstake, The interleukin 23 receptor gene does not confer risk to systemic sclerosis and is not associated with systemic sclerosis disease phenotype, Ann. Rheum. Dis. 68 (2009) 253–256.
[29] B. Farago, L. Magyari, E. Safrany, V. Csongei, L. Jaromi, K. Horvatovich, C. Sipeky, A. Maasz, J. Radics, A. Gyetvai, Z. Szekanecz, L. Czirjak, B. Melegh, Functional variants of interleukin-23 receptor gene confer risk for rheumatoid arthritis but not for systemic sclerosis, Ann. Rheum. Dis. 67 (2008) 248–250.
[30] W. Nakayama, M. Jinnin, Y. Tomizawa, K. Nakamura, H. Kudo, K. Inoue, K. Makino, N. Honda, I. Kajihara, S. Fukushima, H. Ihn, Dysregulated interleukin-23 signaling contributes to the increased collagen production in scleroderma fibroblasts via balancing microRNA expression, Rheumatol. Oxford (Oxford) 56 (2017) 145–155.
[31] K. Komura, M. Fujimoto, M. Hasegawa, F. Ogawa, T. Hara, E. Muroi, K. Takehara, S. Sato, Increased serum interleukin 23 in patients with systemic sclerosis, J. Rheumatol. 35 (2008) 120–125.
[32] J. Jin, C. Chou, M. Lima, D. Zhou, X. Zhou, Systemic sclerosis is a complex disease associated mainly with immune regulatory and inflammatory genes, Open Rheumatol. J. 8 (2014) 29–42.
[33] S.K. Agarwal, P. Gourh, S. Shete, G. Paz, D. Divecha, J.D. Reveille, S. Assassi, F.K. Tan, M.D. Mayes, F.C. Arnett, Association of interleukin 23 receptor polymorphisms with anti-topoisomerase-I positivity and pulmonary hypertension in systemic sclerosis, J. Rheumatol. 36 (2009) 2715–2723.
[34] A. Ichihara, M. Jinnin, H. Ihn, Treatment of psoriasis with ustekinumab improved skin tightening in systemic sclerosis, Clin. Exp. Rheumatol. 35 (Suppl 106) (2017) 208–210.
[35] S. Olivier, M. Foretz, B. Viollet, Promise and challenges for direct small molecule AMPK activators, Biochem. Pharmacol. 153 (2018) 147–158.
[36] I. Tamargo-Gomez, G. Marino, AMPK: regulation of metabolic dynamics in the context of autophagy, Int. J. Mol. Sci. 19 (2018).
[37] J. Gao, J. Ye, Y. Ying, H. Lin, Z. Luo, Negative regulation of TGF-beta by AMPK and implications in the treatment of associated disorders, Acta. Biochim. Biophys. Sin (Shanghai). 50 (2018) 523–531.

[38] C. Beyer, J.H. Distler, Morphogen pathways in systemic sclerosis, Curr. Rheumatol. Rep. 15 (2013) 299.
[39] C. Dees, P. Zerr, M. Tomcik, C. Beyer, A. Horn, A. Akhmetshina, K. Palumbo, N. Reich, J. Zwerina, M. Sticherling, M.P. Mattson, O. Distler, G. Schett, J.H. Distler, Inhibition of Notch signaling prevents experimental fibrosis and induces regression of established fibrosis, Arthritis Rheum. 63 (2011) 1396–1404.
[40] C. Dees, M. Tomcik, P. Zerr, A. Akhmetshina, A. Horn, K. Palumbo, C. Beyer, J. Zwerina, O. Distler, G. Schett, J.H. Distler, Notch signaling regulates fibroblast activation and collagen release in systemic sclerosis, Ann. Rheum. Dis. 70 (2011) 1304–1310.
[41] S. Harrach, V. Barz, T. Pap, H. Pavenstadt, E. Schlatter, B. Edemir, J. Distler, G. Ciarimboli, J. Bertrand, Notch signaling activity determines uptake and biological effect of imatinib in systemic sclerosis dermal fibroblasts, J. Invest. Dermatol. 139 (2019) 439–447.
[42] Y. Wang, Y. Yang, Y. Luo, Y. Yin, Q. Wang, Y. Li, T. Kanekura, J. Wang, G. Liang, M. Zhao, Q. Lu, R. Xiao, Aberrant histone modification in peripheral blood B cells from patients with systemic sclerosis, Clin. Immunol. 149 (2013) 46–54.
[43] J. Henderson, J. Distler, S. O'Reilly, The role of epigenetic modifications in systemic sclerosis: a druggable target, Trends Mol. Med. 25 (2019) 395–411.

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